# AI & SOCIETY COURSE

**Prof. Mor Naaman**

Part 2

# AI AND INDIVIDUALS

Unit 1: How do people use AI?

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## Where We Are in the Course

Part 1 — AI Foundations
1. What is Generative AI: How it Works and What it Can Do
2. Bias and Limitations: Where GenAI Fails

Part 2 — AI and Individuals
3. AI and You: Cognition, Reliance, and Creativity
4. Persuasion by Design: How AI Influences You

Part 3 — AI, Communication, and Information
5. AI-Mediated Communication and Relationships
6. AI, Culture, and the Information Environment

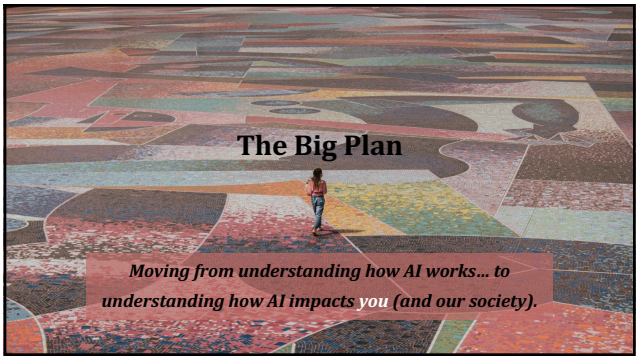
Part 4 — Challenges for AI Implementation
7. Trust and Safety: AI Harms and Prevention
8. Data, Privacy, and IP: Who Owns What
9. The Challenging Economics of AI

Part 5 — AI's World-wide Impact
10. AI and the World: Education, Work, Power, Democracy
11. Externalities: Human and Environmental Costs of AI

Part 6 — The Way Forward with AI
12. The Future: Navigating AI in Your Community



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## Using Science



MY APPROACH

In much of this course, from now on, I am going to use ***scientific knowledge*** from academics around the world (including myself) who are working to understand how AI impacts our society.



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## The plan for Unit 2

**[1] AI and You: Cognition and Creativity**
Goal: Understand how using AI changes the way individuals learn, think and create.

**[2] AI Relationships: Trust, Reliance, Anthropomorphism and Bonding**
Goal: Understand how individual trust, rely, and connect with AI and when they shouldn't.

**[3] Persuasion by Design: How AI Influences You**
Goal: Learn how AI can be designed to change your mind and guide your choices.



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## This lesson plan and goals

- **What are people using AI for?**
- **Understand AI's impact on our thinking and learning**
- **Understand AI's Impact on our creativity**



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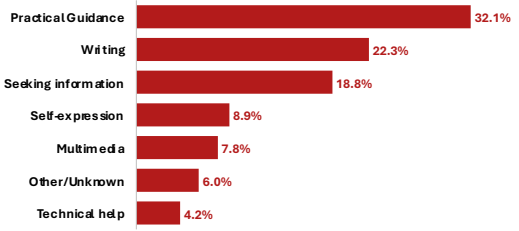
AI in the real world

## What are people using AI chatbots for?

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*



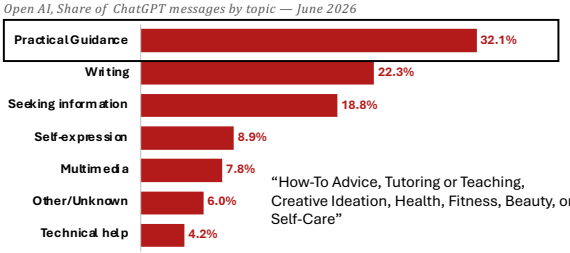

*Source: openai.com/signals/data/*

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*




"How-To Advice, Tutoring or Teaching, Creative Ideation, Health, Fitness, Beauty, or Self-Care"

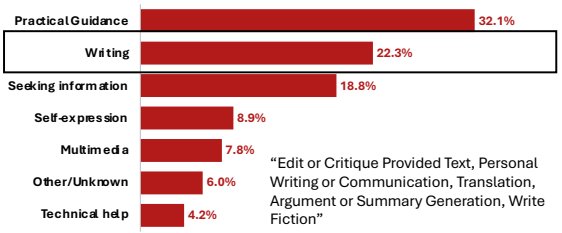
*Source: openai.com/signals/data/*

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*




"Edit or Critique Provided Text, Personal Writing or Communication, Translation, Argument or Summary Generation, Write Fiction"

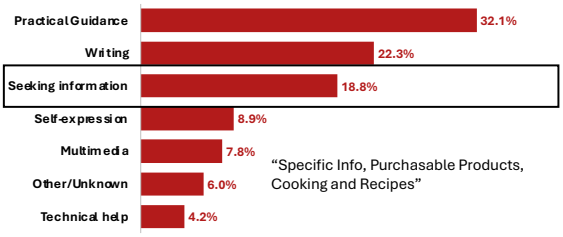
*Source: openai.com/signals/data/*

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*




"Specific Info, Purchasable Products, Cooking and Recipes"

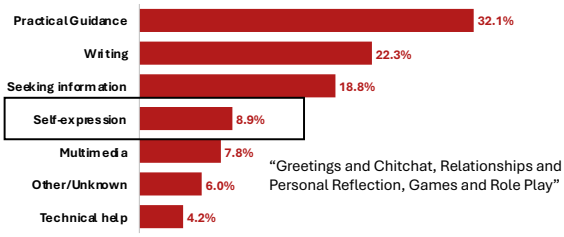
*Source: openai.com/signals/data/*

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*




"Greetings and Chitchat, Relationships and Personal Reflection, Games and Role Play"

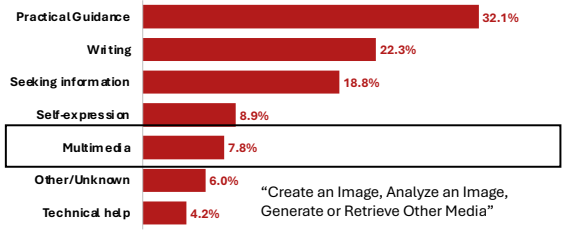
*Source: openai.com/signals/data/*

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*



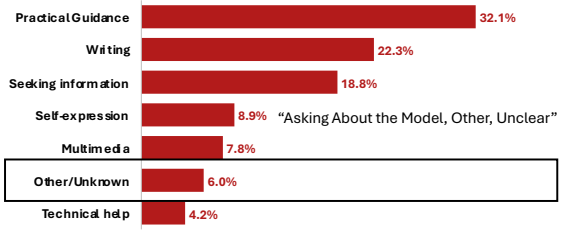

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*



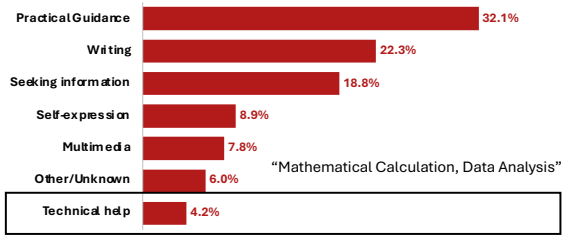

*Source: openai.com/signals/data/*

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*



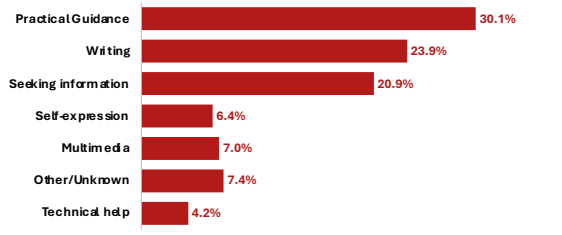

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## What are people using it for? Bangladesh

*Open AI, Share of ChatGPT messages by topic — June 2026*

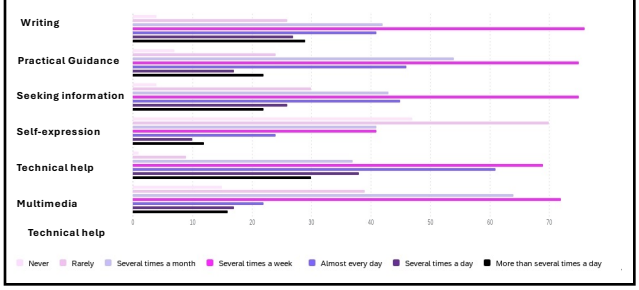



*Source: openai.com/signals/data/*

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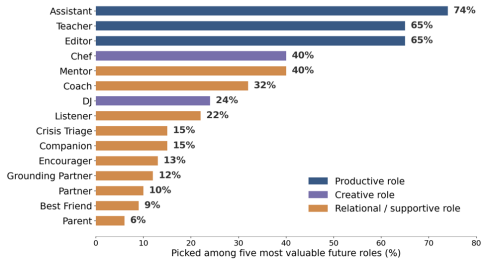
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## What are people using it for? (You)




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## What do people want chatbots to be?



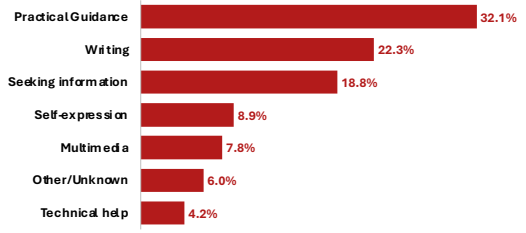

*Source: Inflection AI, "Who's Actually Using Chatbots in 2026?" & "Five Kinds of Chatbot Users" (Jul 2026)*

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## What are people using it for?

*Open AI, Share of ChatGPT messages by topic — June 2026*




*Source: openai.com/signals/data/*

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## People juggle several AI chatbots (survey)

**4.2 bots**
used regularly, on average (median 4.0)

**1.8 bots**
used "several times a day" or "at least once a day"

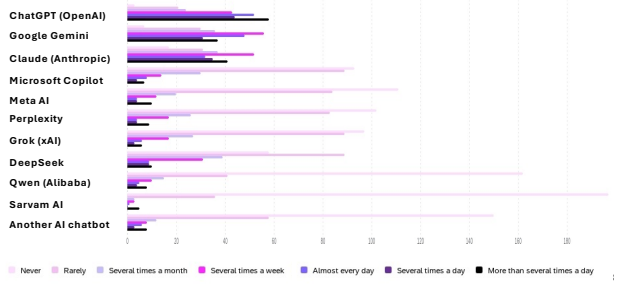
**Most used (1,000 people in survey):**
**ChatGPT 91% · Gemini 64% · Siri 45% · Claude 29%**

*Source: Inflection AI, "Who's Actually Using Chatbots in 2026?" & "Five Kinds of Chatbot Users" (Jul 2026)*

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## Using chatbots (you)




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## Does AI help people get stuff done?

**Agency** *(n.)*
The ability of people to take meaningful action to shape the world around them.

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## People gain agency from AI

Can complete tasks efficiently

*"I took a photo with ChatGPT, and it totally understood. That was a big, 'hell yeah' moment."*
—Liam, U.S.

*Source: Beacock et al., "AI Usage Patterns Are Shaped by Perceived Gains in Human Agency" (arXiv:2607.02313, 2026)*

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## People gain agency from AI

Can think more clearly, organize thoughts, develop opinions.

*"I have a lot of deep conversations with AI about philosophical topics and it helps me structure my thinking."*
—Lazar, Germany

*Source: Beacock et al., "AI Usage Patterns Are Shaped by Perceived Gains in Human Agency" (arXiv:2607.02313, 2026)*

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## People gain agency from AI

Can understand and process emotional states.

*"[AI] gives me enough judgment to help me get to that level of clarity for myself."*

—Angela, U.S.

*Source: Beacock et al., "AI Usage Patterns Are Shaped by Perceived Gains in Human Agency" (arXiv:2607.02313, 2026)*

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## People gain agency from AI

Can better manage social connections

*"You can [use AI agents to] analyze a LinkedIn profile… game out what the interview is going to look like."*

—Teddy, U.S.

*Source: Beacock et al., "AI Usage Patterns Are Shaped by Perceived Gains in Human Agency" (arXiv:2607.02313, 2026)*

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## People gain agency from AI

Can better handle complex systems and institutions.

*"Now I can test and see if [[doctors] are] worth their expertise… I no longer feel that I'm out of control."*

—Koh, Singapore

*Source: Beacock et al., "AI Usage Patterns Are Shaped by Perceived Gains in Human Agency" (arXiv:2607.02313, 2026)*

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## But…

**It is possible that:**

- **People think they are doing more, but it is not real**
- **They do more, but no material benefits or new skills.**

*Source: Beacock et al., "AI Usage Patterns Are Shaped by Perceived Gains in Human Agency" (arXiv:2607.02313, 2026)*

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## Given all the different ways we use it…

**… How does AI impact us as individuals?**

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## At least 40%

**of ChatGPT messages worldwide have AI write for us, create arguments for us, help us generate ideas.**

**What happens to our cognition?**

*Source: openai.com/signals/data/*

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**When AI is generating ideas, making images and videos, creating content for us**

**What Happens to Creativity?**

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## At least 58%

**of work-related messages ask AI to give us information or summarize information for us**

**How do we trust it and rely on its answers?**

*Source: Counts et al., "AI in the Enterprise" (arXiv:2605.23958, 2026)*

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## 9% "Self Expression"

**"Self-Expression" messages include greetings, personal reflection, and role play**

**What happens when we treat AI like a person?**

*Source: openai.com/signals/data/*

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## 80%

**of chatbot users want some emotional role from AI in the future — listener, coach, friend, companion**

**What are the consequences of AI relationships?**

*Source: Inflection AI, "Five Kinds of Chatbot Users," 2026*

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Our brains with AI

## AI's Impact on our Thinking and Learning

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## This lesson plan

- What are people using AI for?
- Impact on our thinking and learning
- Impact on our creativity

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## Evidence from Science

**High-school students learning math did better on practice problems with a ChatGPT tutor… but worse on the exam without it**

| Practice | Exam |
|---|---|
| ▲ ChatGPT better than control | ▼ ChatGPT worse than control |

*Slide credit: Prof. Andrew Perfors*
*Source: Bastani et al. (2025), "Generative AI without guardrails can harm learning: Evidence from high school mathematics"*

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## Evidence from Science

**High-school students learning math did better on practice problems with a ChatGPT tutor… but worse on the exam without it**

| Practice | Exam |
|---|---|
| ▲ ChatGPT better than control | ▼ ChatGPT worse than control |

**Also: students who used ChatGPT *thought* they performed better on the exam than those who didn't!**

*Slide credit: Prof. Andrew Perfors*
*Source: Bastani et al. (2025), "Generative AI without guardrails can harm learning: Evidence from high school mathematics"*

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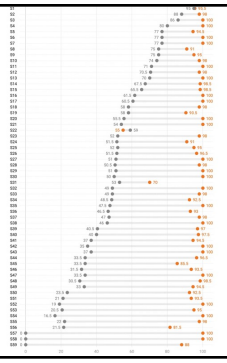



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## One professor's class

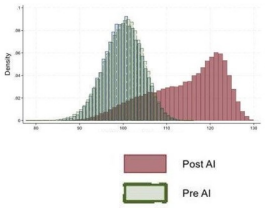
**Brown University**
**ECON 1170 course**

midterm (with AI)
final (without)



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## The Generative AI Learning Penalty




Study of 26,000 students in China measured casual impact of AI use on homework and exam scores

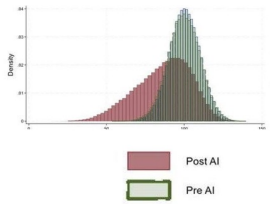
- ▲ Homework scores 18%
- ▼ Completion time 30%
- ▼ Monthly exam scores fell 20%
- ▼ high-stakes entrance-exam scores dropped 18–24%.

*Source: Strömberg et al. (2026), "The Generative AI Learning Penalty: Evidence from Chinese Secondary Education"*

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## The Generative AI Learning Penalty




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*Source: Strömberg et al. (2026), "The Generative AI Learning Penalty: Evidence from Chinese Secondary Education"*

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## But: AI can help under certain conditions

"Generative AI boosts productivity, but its effect on learning new skills is less clear. This large-scale field experiment in a high school found that AI tutoring (GPT-4) improved performance during practice — but students who relied on it underperformed once AI access was removed, suggesting weaker skill acquisition. **Carefully designed safeguards — especially having the AI give teacher-designed hints instead of answers — largely prevented this harm.**"

-- Bastani et al.

*Source: Bastani et al. (2025), "Generative AI without guardrails can harm learning: Evidence from high school mathematics*

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## Cognitive Offload and False Confidence

People who used ChatGPT instead of web search to make an evidence-based recommendation…

Students researched the safety of sunscreen, then gave a recommendation with justification to a friend.

One group used **ChatGPT**, and another group used **Google Search**.

*Slide credit: Prof. Andrew Perfors*
*Source: Stadler et al. (2024), "Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry"*

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## Cognitive Offload and False Confidence

People who used ChatGPT instead of web search to make an evidence-based recommendation **thought the task was easier… but performed worse**

Students researched the safety of sunscreen, then gave a recommendation with justification to a friend.

One group used **ChatGPT**, and another group used **Google Search**.

▼ **ChatGPT group reported lower cognitive load**
*(less likely to agree with statements like "this task was very complex")*

▼ **ChatGPT group gave lower-quality recommendations**
*(rated independently on accuracy and comprehensiveness)*

*Slide credit: Prof. Andrew Perfors*
*Source: Stadler et al. (2024), "Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry"*

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## Reactive Writing / Mental Hijacking

**Danger: AI provides easy answers, prevents you from developing your own ideas**

My own research: participants were asked to write an essay about an important issue with or without AI help ("auto complete").

- **Participants were led by the AI suggestions, reacting to it instead of developing their own ideas**
- **Were also influenced in other ways (more on that later)**

*Source: Bhat et al. (2026), Reactive Writers: How Co-Writing with AI Changes How We Engage with Ideas*

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## Depth, Synthesis, and Integration

People who learn about a topic via LLMs compared to learning via Web Search…

People were asked to research and give advice to a friend on how to plant a vegetable garden.

One group used **ChatGPT**, and another group used **Google Search.**

*Slide credit: Prof. Andrew Perfors*
*Source: Melumad & Yun (2025), "Experimental evidence of the effects of large language models versus web search on depth of learning"*

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## Depth, Synthesis, and Integration

People who learn about a topic via LLMs compared to learning via Web Search **developed shallower knowledge** (even when LLMs provide links).

People were asked to research and give advice to a friend on how to plant a vegetable garden.

One group used **ChatGPT**, and another group used **Google Search.**

**ChatGPT group:**

- ▼ **developed shallower knowledge**
- ▼ **put less effort in forming advice**
- ▼ **wrote shorter advice**
- ▼ **wrote advice that others liked less**

**… then the people who used search**

*Slide credit: Prof. Andrew Perfors*

*Source: Melumad & Yun (2025), "Experimental evidence of the effects of large language models versus web search on depth of learning"*

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## How is Using AI Impacting Human Learning?

**LLMs help learning if they:**

- Give real practice on the skills that matter
- Offer accurate feedback that corrects mistakes
- Support learners without doing the work for them
- Encourage deeper thinking and connecting ideas

*Slide credit: Prof. Andrew Perfors*

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## How is Using AI Impacting Human Learning?

**LLMs help learning if they:**

- Give real practice on the skills that matter
- Offer accurate feedback that corrects mistakes
- Support learners without doing the work for them
- Encourage deeper thinking and connecting ideas

**LLMs hurt learning if they:**

- Skip the practice that builds basic skills
- Interrupt the thinking that builds understanding
- Block the knowledge needed to catch errors
- Remove motivation and a sense of ownership

*Slide credit: Prof. Andrew Perfors*

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## Summary so far

- AI can boost short-term performance and confidence
- But looks like there is a cost to real learning and critical thinking.
- We don't yet know the long-term effects
- It may depend on how AI is used!

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Our brains with AI

## AI's Impact on our Thinking and Learning

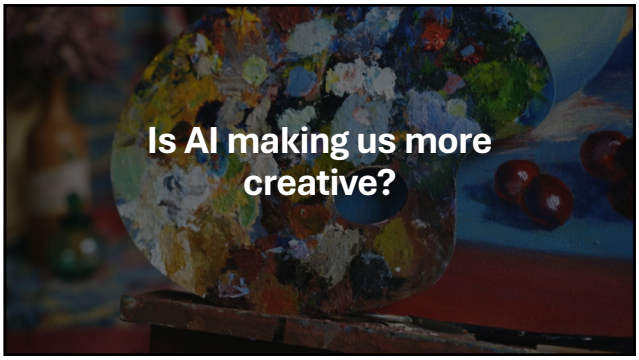
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## This lesson plan

- What are people using AI for?
- Impact on our thinking and learning
- **Impact on our creativity**

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## Exercise: AI Short Stories

- **Go to your favorite AI chatbot. Enter the prompt:**
- "in about 100 words, write a complete short story about Bangladesh, in English, with a clear beginning, middle, and end. Focus on one main character, one moment of change, and use specific details."

Submit your story to the "AI writes a short story" link at ai-course.mornaaman.com

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## Study 1: Essay writing, humans vs. AI

- In the US, students submit creative essays for college admissions
- The study compared human vs. GPT-4 college admissions essays
- Measured: how many new ideas each additional essay adds to the group
- Each additional human essay more new ideas than each GPT-4 essay
- The gap widened as more essays were added

*Moon, Green & Kushlev (2024), "Homogenizing effect of LLMs on creative diversity"*

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## Study 2: Essay writing w/ AI

- Compared human, humans with one AI idea, humans with five AI ideas
- Evaluators rated the AI users' stories as **more novel** and **more useful** than stories written by people who could not use ChatGPT
- But people's stories also became more similar to each other's when they used AI ideas

**Widespread LLM use could reduce the collective diversity of creative ideas, even as it lifts individual writers**

*Doshi & Hauser (2024). "Generative AI enhances individual creativity but reduces the collective diversity of novel content"*

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## Study 3: brainstorming

- Across 5 experiments, brainstorming sessions assisted by ChatGPT consistently produced narrower sets of ideas than unassisted human brainstorming
- Example task: invent a toy using only a brick and a fan
  - *94% of ChatGPT-assisted ideas shared overlapping concepts*
  - *For example: 9 different participants independently named their toy "Build-a-Breeze Castle"*
  - *Human-only ideas were entirely unique from one another*

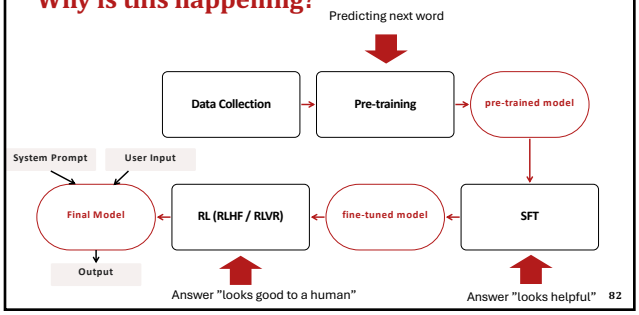
**ChatGPT can improve an individual idea -- while limiting what the group comes up with together**

*Meincke, L., Nave, G., Terwiesch, C. (2025). "ChatGPT decreases idea diversity in brainstorming."*

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## Why is this happening?




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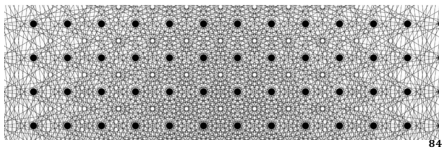
## But at the same time...

Technology > Artificial Intelligence

**OpenAI's internal AI model just solved an 80-year-old math problem — and mathematicians verified it**

The closest the field has come to solving the planar unit distance problem, first proposed in the 1940s, was in 1984. Now, OpenAI claims an internal model has cracked the puzzle.

By Drew Turney | Published May 29, 2026 in News



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## But at the same time...

CNN Business — Subscribe

BUSINESS • 4 MIN READ

**OpenAI says it has solved one of math's 'Millennium Problems'**

SEP 9, 2026

By Karina Tsui

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## Why?

If GenAI enhances individual creativity but reduces the collective diversity, how is it able to solve all these math problems?

- Math is a task with one correct answer, verifiable. There is a "good answer"; learns (RLVR) to explore
- Creative tasks value comes from variance, diversity, not focus on "best path"
- Leverages huge "explore" capacity to try again and again in math; essay writing harder to repeat

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## What to do to be creative with AI?

- **Generate your own idea first, then bring in AI**
- **Push AI for unusual answers**
- **Treat AI as one voice among many**
- **Vary your prompts (and tools)**

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## Quick Summary

- AI can make each of us individually more creative
- But when many people use it for the task, the pool of ideas gets more similar
- We have seen: essays, brainstorming; also happens in other domains (academic writing!).
- **Best use: generate your own idea first, then bring in AI to expand, critique, or vary it**

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## Overall Summary

- Some key ways people use AI: practical help, writing, and finding information
- People increasingly want AI to play emotional roles: coach, listener, friend.
- AI can boost short-term performance and confidence, often at a cost to real learning and critical thinking.
- Mental hijacking: AI's quick answers/suggestions can impact how we think
- AI can help creativity for one person, but may reduce overall creativity

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